Amir-Reza Negari

Postdoctoral Fellow | Institute for Quantum Computing, University of Waterloo
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Physics Ph.D. with research in quantum information, error correction, and statistical physics, combining analytical theory with numerical methods. Six peer-reviewed papers and seven preprints.

Technical Skills

Python: JAX, PyTorch, CuPy, Numba CUDA.

Additional languages: C/C++, CUDA, Julia, MATLAB.

Scientific computing: GPU/CPU parallel programming; distributed and high-performance computing; Slurm.

Methods: Tensor networks (DMRG/MPS), determinantal quantum Monte Carlo, Gaussian methods, stabilizer simulation, and quantum error-correction decoders.

Academic Background

Postdoctoral Fellow, Institute for Quantum Computing 2026–present
University of Waterloo, Quantum Interactions Theory Group.

Ph.D. in Physics, Perimeter Institute for Theoretical Physics 2022–2026
Supervisor: Timothy H. Hsieh.
Thesis: *Phases of matter in quantum information and error correction*.

M.Sc. in Physics, Perimeter Institute for Theoretical Physics 2021–2022
Perimeter Scholars International.

B.Sc., Sharif University of Technology 2016–2021

Publications and Preprints

Ordered by relevance to AI research and scientific computing.

1. **Unlearnable phases of matter**
   Tarun Advaith Kumar, Yijian Zou, **Amir-Reza Negari**, Roger G. Melko, Timothy H. Hsieh.
   Preprint (2026). arXiv:2602.11262
2. **Amelioration for the Sign Problem: An Adiabatic Quantum Monte Carlo Algorithm**
   Mohammad-Sadegh Vaezi, **Amir-Reza Negari**, Amin Moharramipour, Abolhassan Vaezi.
   Physical Review Letters **127**, 217003 (2021). arXiv:2012.12911
3. **Extendibility of Fermionic Gaussian States**
   Amir-Reza Negari, Farzin Salek.
   Physical Review Letters **137**, 080201 (2026). arXiv:2508.18532
4. **Chiral Ising Gross–Neveu Criticality of a Single Dirac Cone: A Quantum Monte Carlo Study**
   S. Mojtaba Tabatabaei*, **Amir-Reza Negari***, Joseph Maciejko, Abolhassan Vaezi.
   Physical Review Letters **128**, 225701 (2022). *Editors' Suggestion*. arXiv:2112.09209
5. **Spacetime Markov length: A diagnostic for fault tolerance via mixed-state phases**
   Amir-Reza Negari, Tyler D. Ellison, Timothy H. Hsieh.
   Physical Review B **114**, 045130 (2026). arXiv:2412.00193
6. **Critical non-equilibrium phases from noisy topological memories**
   Amir-Reza Negari, Subhayan Sahu, Jan Behrends, Benjamin Béri, Timothy H. Hsieh.
   Preprint (2026). arXiv:2601.10792

Publications and Preprints (continued)

7. **Instability of Markov chains under infinitesimal coherent dissipation**
   Amir-Reza Negari, Yijian Zou, Timothy H. Hsieh.
   Preprint (2026). arXiv:2610.18983
8. **Approximation theorems for fermionic Gaussian states**
   Amir-Reza Negari, Farzin Salek, Zoltán Zimborás, Aram Harrow, Patrick Hayden, Jens Eisert.
   Preprint (2026). arXiv:2610.01860
9. **Measurement-induced phase transitions in the toric code**
   Amir-Reza Negari, Subhayan Sahu, Timothy H. Hsieh.
   Physical Review B **109**, 125148 (2024). arXiv:2307.02292
10. **Nonadditivity in fermionic quantum Shannon theory**
    Farzin Salek, **Amir-Reza Negari**, Graeme Smith, Jens Eisert, Debbie Leung.
    Preprint (2026). arXiv:2610.06758
11. **Entanglement of purification for Werner states: canonical purification and nonadditivity**
    Zahra Baghali Khanian, **Amir-Reza Negari**.
    Preprint (2026). arXiv:2610.03630
12. **Rényi Entanglement of Purification Is Non-additive**
    Amir-Reza Negari, Zahra Baghali Khanian.
    Preprint (2026). arXiv:2608.28897
13. **Symmetry enforcing entanglement at high temperatures**
    Amir-Reza Negari[*], Leonardo A. Lessa[*], Subhayan Sahu.
    Physical Review A **114**, 022458 (2026). arXiv:2508.20166

[*] Equal contribution.

Selected Awards

- **Bronze Medal**, International Physics Olympiad, Zurich 2016
- **Gold Medal**, Iran's National Physics Olympiad 2015
- **Scholarships and fellowships:** Institute for Quantum Computing fellowship; University of Waterloo international student scholarship and Perimeter Residence Scholarship (2022–2026); Perimeter International Scholarship (2021–2022).

Mentoring, Leadership, Service, and Presentations

- **Mentoring:** Mentored a Perimeter Scholars International student (2023–2025); supervised a high-school research group through Perimeter's ISSYP program.
- **Research community:** Founded the Eureka Club and a biweekly Quantum Matter arXiv lunch; co-founded the Topological Quantum Club at Perimeter.
- **Peer review:** QIP conference (2026), *Physical Review B*, and *PRX Quantum*.
- **Teaching:** Graduate and undergraduate physics teaching assistantships (2018–2022); Physics Olympiad instruction and outreach (2015–2019).
- **Presentations:** Critical Mixed-State Phases from Coherent Errors: CMI, Loop Constraints, and Network Conductance (talk, University of Cologne, online, 2025); Extendibility of Fermionic Gaussian States (contributed talk, Year of Quantum Across Canada, Institute for Quantum Computing, 2025); Fault tolerance via mixed-state phases (talk, Perimeter Institute, 2025); Mixed-state quantum criticality (poster, QUDYN25, Max Planck Institute, 2025); Measurement-induced phase transitions in the toric code (talk, Perimeter Institute, 2023); Classical simulability of MBQC and measurement-induced transitions in the toric code (invited talk, University of British Columbia, 2023); Measurement-induced phase transitions in the toric code (poster, Boulder Summer School, 2023).